PaperSonar

Business questions

Design baseline 1.0 / 25 October 2026

The following questions define PaperSonar's analytical workload. Each question names the decision, measure, warehouse path and interpretation limit. The first ten form the minimum reporting set. Questions eleven to fourteen extend citation, quality and sensitivity analysis.

Q01 Which selected topics grew most between 2023 and 2025

Decision. Choose a direction for a deeper review.

Measure. Distinct admitted works by primary topic and publication year. Compare 2023 and 2025 counts and a two-year CAGR where both counts are positive.

Data. Publication fact, work and topic dimensions.

Analysis. Compare topic > subfield > field and year. Show counts next to percentages.

Limit. Suppress growth rankings when either endpoint has fewer than 30 works. This is a proposed reporting threshold.

Q02 Which topics are expanding their share of the retained corpus

Decision. Separate broad corpus growth from a change in relative attention.

Measure. Topic primary-publication count divided by all admitted publications in the same year. Report percentage-point change from 2023 to 2025.

Data. Publication fact and topic dimension.

Analysis. Slice by year and work type, then rank topics.

Limit. Shares describe the selected corpus, not all research worldwide.

Q03 Which topics have high citation impact within comparable publication cohorts

Decision. Prioritize a reading shortlist without favoring older work.

Measure. At one snapshot date, median cited_by_count within publication year and work type. Report cohort size and missing counts.

Data. Publication and snapshot facts joined by work_key.

Analysis. Dice by topic, publication year and type.

Limit. Use one observation date. Citation counts do not establish research quality or causation.

Business questions continued

Q04 Which publications gained citations between two observations

Decision. Monitor papers that merit a fresh review.

Measure. Difference between the same work's citation counts at two successfully observed dates. Rank positive changes and display negative revisions separately.

Data. Work snapshot fact with role-played date dimension.

Analysis. Drill to publication title and DOI.

Limit. Only works present at both dates qualify. Snapshot differences begin with local collection, not publication.

Q05 Which topic intersections are sparsely represented but supported by substantial parent topics

Decision. Identify candidates for manual literature review.

Measure. For each unordered pair of supplied topics count distinct co-assigned works. Display each parent count, joint count and joint / smaller parent count for 2021 to 2025.

Data. Work topic bridge with publication and topic dimensions.

Analysis. Compare pairs and inspect their actual papers.

Limit. Require each parent to have at least 100 works and the pair to have at least 5. Sparse tagging is a prompt for review, not proof of a scientific gap.

Q06 Which venues regularly publish the selected topics

Decision. Build a venue reading list.

Measure. Count distinct admitted works by primary venue and primary topic for 2021 to 2025. Display missing venue share.

Data. Publication fact, venue and topic dimensions.

Analysis. Drill topic to venue to work.

Limit. Primary source is one indexed location and may be a repository. This does not rank venue quality.

Business questions continued

Q07 Which institutions contribute to each topic

Decision. Identify potential collaborators.

Measure. Sum institution allocation weights by primary topic and year. Show distinct full publication counts alongside fractional credit.

Data. Publication fact and work institution bridge.

Analysis. Roll institution to country and compare years.

Limit. Deduplicate institutions across authors before assigning weights. Unknown affiliations form a visible coverage category.

Q08 How concentrated is institutional participation

Decision. Find topics dominated by a small set of institutions.

Measure. Largest five institutions' fractional credit divided by all fractional institution credit within each primary topic.

Data. Publication fact and work institution bridge.

Analysis. Dice by year, topic and country.

Limit. Compute the top five after filtering. Exclude unknown institutions from ranking and disclose their publication share.

Q09 How does open access availability vary by topic and year

Decision. Estimate how accessible a reading direction will be.

Measure. At one snapshot date divide observed open-access works by works with known access status. Display unknown-status share separately.

Data. Publication and snapshot facts with access dimension.

Analysis. Slice by primary topic, year and work type.

Limit. Availability reflects the observation date, not necessarily availability on publication day.

Business questions continued

Q10 Which authors have repeated activity in a selected direction

Decision. Prepare an expert shortlist for manual review.

Measure. Count distinct admitted works per source author identifier over 2021 to 2025, with latest publication year.

Data. Work author bridge, author dimension and publication fact.

Analysis. Drill topic to author to work.

Limit. Names do not identify people reliably. Use OpenAlex ID and ORCID where supplied. This is activity, not an expertise certification.

Q11 Which selected topic communities cite one another

Decision. Find intellectual links that a title search may miss.

Measure. Count current citation pairs with both endpoints admitted, grouped by citing and cited primary topic. Keep direction.

Data. Citation fact joined twice to publication fact and topic dimension.

Analysis. Filter citing year and inspect edge-level evidence.

Limit. This is an induced subgraph. It cannot reproduce global citation totals.

Q12 Which comparisons are weakened by missing or unstable metadata

Decision. Avoid presenting unreliable rankings.

Measure. Missing DOI, primary topic, venue, institution and known access percentages over their eligible work populations. Compare corpus versions through archived outputs.

Data. Work dimension, publication fact, bridges and snapshot fact.

Analysis. Drill from a flagged topic to the affected records.

Limit. Do not turn nulls into zero citation counts. Failed refreshes must not appear as changes.

Business questions continued

Q13 How sensitive is an opportunity shortlist to the counting method

Decision. See whether a recommendation survives reasonable analytical choices.

Measure. Compare rankings using primary-topic counts and equal fractional topic allocation, with and without low-volume groups.

Data. Publication fact and work topic bridge.

Analysis. Inspect rank changes and supporting work lists.

Limit. Differences describe method sensitivity. No automatic confidence score or discovery claim is assigned.

Q14 How does the current partial year compare with the same elapsed period last year

Decision. Monitor recent activity without comparing a partial year to a complete one.

Measure. Count works through a common month and day cutoff in 2025 and 2026 using source-reported dates. Show cutoff and retrieval date.

Data. Publication fact with date and work dimensions.

Analysis. Slice by topic and type, then drill by month.

Limit. Exclude unknown or year-only dates from elapsed-period comparisons. Indexing delay remains a limitation.